

Conduction Heat Transfer Tutorial

1 Questions

1. One face of a copper plate $10\text{ cm} \times 25\text{ cm} \times 3\text{ cm}$ thick is maintained at $400\text{ }^{\circ}\text{C}$. The other face is maintained at $100\text{ }^{\circ}\text{C}$. Calculate the heat transfer rate through the plate (Ans. 92.5 kW).
2. An exterior of a house may be approximated by a 10.2 cm layer of common brick [$k = 0.7\text{ W/m }^{\circ}\text{C}$] followed by a 3.8 cm layer of gypsum [$k = 0.48\text{ W/m }^{\circ}\text{C}$]. What thickness of loosely packed rock wool insulation [$k = 0.065\text{ W/m }^{\circ}\text{C}$] should be added to reduce the heat loss (or gain) through the wall by 80% ? [$Q_{\text{with insulation}} / Q_{\text{without insulation}} = 0.2$]. (Ans. 0.0584 m).
3. A thick walled tube of stainless steel [$k = 19\text{ W/m }^{\circ}\text{C}$] with 2 cm inner diameter and 4 cm outer diameter is covered with a 3 cm layer of asbestos insulation [$k = 0.2\text{ W/m }^{\circ}\text{C}$]. The inside wall temperature of the pipe is maintained at $600\text{ }^{\circ}\text{C}$, and outside surface temperature is $100\text{ }^{\circ}\text{C}$. Calculate the heat loss per meter of length. Also calculate the tube insulation interface temperature. (Ans. 680 W/m , $595.8\text{ }^{\circ}\text{C}$)
4. Calculate the critical radius of insulation for asbestos [$k = 0.17\text{ W/mK}$] surrounding a pipe and exposed to room air at $20\text{ }^{\circ}\text{C}$ with $h = 3\text{ W/m}^2\text{K}$. Calculate the heat loss from a $200\text{ }^{\circ}\text{C}$, 5 cm diameter pipe when covered with the critical radius of insulation and without insulation. [Ans. $Q_{\text{criti}} = 105.7\text{ W/m}$, without 85 W/m]
5. A slab of thickness 4 cm is subjected to a hot air flow at $130\text{ }^{\circ}\text{C}$ with $h = 500\text{ W/m}^2\text{K}$ and cold air at $30\text{ }^{\circ}\text{C}$ with $h = 250\text{ W/m}^2\text{K}$ on each side. The plate thermal conductivity is $20\text{ W/m }^{\circ}\text{C}$. Estimate heat transfer through the plate per meter square. [Ans 12.5 kW]
6. A current of 200 A is passed through a stainless steel wire [$k = 19\text{ W/mK}$] 3 mm in diameter. The resistivity of steel may be assumed as $70\text{ }\mu\Omega\text{ cm}$, and the length of the wire is 1 m . The wire is submerged in a liquid at $110\text{ }^{\circ}\text{C}$ and experience a convection heat transfer coefficient of $4\text{ kW/m}^2\text{ }^{\circ}\text{C}$. Calculate the centre temperature of the wire. (Ans $231.6\text{ }^{\circ}\text{C}$)

7. An aluminum rectangular fin [$k = 200 \text{ W/m}^\circ\text{C}$] 3 mm thick and 7.5 cm long protrudes from a wall. The base is maintained at 300°C . And the ambient temperature is 50°C , with $h = 10 \text{ W/m}^2\text{K}$. Calculate the heat loss from the fin per unit depth of material. [Ans. 360 W/m]
8. Aluminum fins 1.5 cm wide and 1 mm thick are placed on a 2.5 cm diameter tube to dissipate the heat. The tube surface is 170°C , and the ambient fluid temperature is 25°C . Calculate the heat loss per fin for $h = 130 \text{ W/m}^2\text{K}$. Assume $k = 200 \text{ W/mK}$ for aluminum. [ans. 61 W].
9. A steel ball [$c = 0.46 \text{ kJ/kg}^\circ\text{C}$, $k = 35 \text{ W/m}^\circ\text{C}$] 5.0 cm in diameter and initially at a uniform temperature of 450°C is suddenly placed in a controlled environment in which the temperature is maintained at 100°C . The convection heat-transfer coefficient is $10 \text{ W/m}^2^\circ\text{C}$. Calculate the time required for the ball to attain a temperature of 150°C .
10. A wall 2 cm thick and thermal conductivity 1.3 W/mK is to be insulated with an average thermal conductivity 0.35 W/mK . Find the thickness of the insulation so that the heat loss per square meter will not exceed 1830 W . Assume the inner and outer surface temperatures are 1300°C , and 30°C , calculate the thickness of the insulation.
11. A certain material 2.5 cm thick, with a cross-sectional area of 0.1 m^2 , has one side at 35°C , and the other at 95°C . The centre is at 62°C , and the heat flow through the material is 1 kW . Obtain an expression for the thermal conductivity of the material as a function of temperature.
12. One side of a copper block 5 cm thick is at 250°C , The other side is covered with a layer of fibre glass 2.5 cm thick. The outside of the fibre glass is maintained at 35°C , and the total heat flow through the slab is 52 kW . What is the area of the slab.
13. A plane wall is with a material having thermal conductivity that varies as the square of the temperature according to the relation $k = k_0(1 + \beta T^2)$. Derive an expression for the heat transfer in such a wall.
14. A material has a thickness of 30 cm and a thermal conductivity of 0.04 W/mK . At a particular instant in time the temperature distribution with x , the distance from the left face, is $T = 150x^2 - 30x$, where x is in meters. Calculate the heat flow rate at $x = 0$, and $x = 30 \text{ cm}$. Is the solid heating up or cooling down ?
15. In a quenching process steel rods ($\rho = 7832 \text{ kg/m}^3$, $c_p = 434 \text{ J/kgK}$, $k = 63.9 \text{ W/mK}$) are heated in a furnace to 850°C and then cooled in a water bath to an average temperature of 95°C . The water bath has a uniform temperature of 40°C and convection heat transfer coefficient of $450 \text{ W/m}^2\text{K}$. If the steel rods have a diameter of 50 mm and length of 2m, determine the time required to cool a steel rod from 850°C to 95°C .

°C in the water bath and total amount of heat transfer during this. (Bi 0.088, $t = 254$ s, $Q = 10.1$ MJ).

16. Estimate the minimum depth at which one must place a water main below the surface to avoid freezing. The soil is initially at a uniform temperature of 20°C . Assume that under the worst conditions anticipated it is subjected to a surface temperature of -15°C for a period of 60 days. Use the following properties for soil. ($\rho = 2050\text{ kg/m}^3$, $k = 0.52\text{ W/mK}$, $c_p = 1840\text{ J/kgK}$, $\alpha = 0.138 \times 10^{-6}\text{ m}^2/\text{s}$) [Ans 0.68 m]
17. A large block of steel [$k = 45\text{ W/mK}$, $\alpha = 1.4 \times 10^{-5}\text{ m}^2/\text{s}$] is initially at a uniform temperature of 35°C . The surface is exposed to a heat flux by suddenly raising the surface temperature to 250°C . Calculate the temperature at a depth of 2.5 cm after a time of 0.5 min. (Ans. 118.5°C)
18. A horizontal pipe 15cm diameter and 4 m long is buried in the earth at a depth of 20 cm. The pipe wall temperature is 75°C , and the earth surface temperature is 5°C . Assuming that the thermal conductivity of the earth is 0.8 W/mK , calculate the heat lost by the pipe. (Ans. 860 W)
19. A small cubical furnace 50 cm by 50 cm by 50cm on the inside is constructed of fireclay brick [$k = 1.04\text{ W/mK}$] with a wall thickness of 10 cm. The inside of the furnace is maintained at 500°C , and the outside is at 50°C . Calculate the heat lost through the walls. [Ans. 8.59 kW]